The Deep Ritz Method

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Outline

1. Introduction
2. Some Machine Learning
3. The Deep Ritz Method
4. Theorems & Experiments

Solving PDEs

- **Finite Difference Method**: Approximate derivatives on a grid.
- **Finite Element Method**: Divide the domain into elements and use variational methods.

Comparison of Numerical Methods

Feature	**FEM**	**FDM**	**DRM**
Higher Dimensions	Limited	Limited	Efficient
Adaptivity	Limited	Limited	Naturally adaptive
Boundary Conditions	Direct enforcement	Direct enforcement	Hard to enforce
Efficiency	Moderate	Moderate	High on GPUs
Acc. w fewer parameters	Moderate	Moderate	High

Blocks

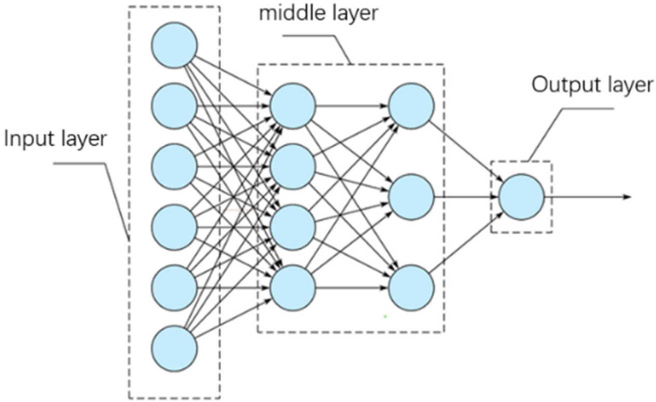

Figure: www.mdpi.com

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Blocks

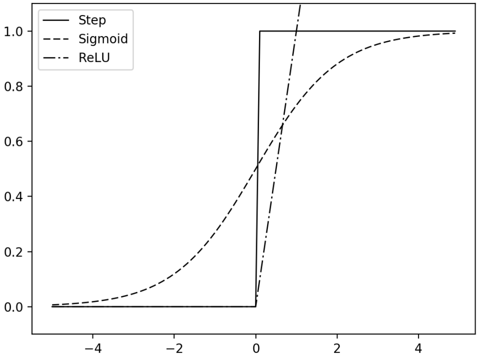
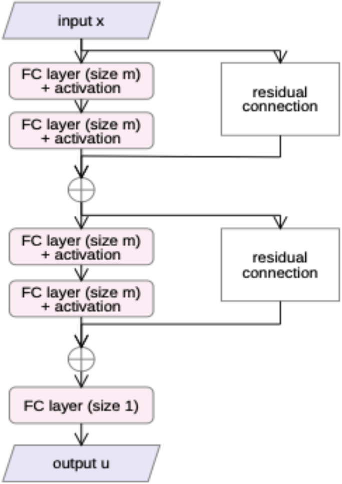


Figure: schwalbe10.github.io

Residual networks



Value function

Score function $L(\theta)$ where θ are the parameters of the model. The goal of training is to find parameters θ^* that minimize this loss function.

Minimalization

- **Gradient Descent (GD):** Iteratively update parameters in the opposite direction of the gradient of the loss function:

$$\theta_{k+1} = \theta_k - \eta \nabla_\theta L(\theta_k)$$

  where η is the learning rate.
- **Backpropagation:** Algorithm used to efficiently compute the gradients $\nabla_\theta L(\theta_k)$ of the loss function with respect to the network parameters θ.

Universal representation

- The Sobolev norm:

$$\|u\|_{H^1(\Omega)} = \left(\int_\Omega |u(x)|^2 dx + \int_\Omega |\nabla u(x)|^2 dx \right)^{1/2}$$

  The space $H_0^1(\Omega)$ refers to functions in $H^1(\Omega)$ that are zero on the boundary $\partial\Omega$.

- Let $\Omega \subseteq \mathbb{R}^d$ be an open set and $u \in H_0^1(\Omega)$. Then for any $\epsilon > 0$, there exists a function $u_\epsilon \in H_0^1(\Omega)$ that can be expressed by a ReLU network of logarithmic depth on d such that:

$$\|u - u_\epsilon\|_{H^1(\Omega)} < \epsilon$$

Problems of Interest

- **Poisson's Equation**: $-\Delta u = f$ in Ω, with $u = 0$ on $\partial\Omega$.
- **Schrödinger Equation**: $-\Delta u + Vu = f$ in Ω, with appropriate boundary conditions.
- **Generalized Poisson's Equation**: -div$\cdot\,(a(x)\nabla u(x)) + b(x)u(x) = f(x)$

Variational Formulation

- **Principle**:
  - Convert PDEs into equivalent(?) minimization problems.
- **Example**:
  - For Poisson's equation, find $u \in H_0^1(\Omega)$ minimizing:

$$E(u) = \frac{1}{2}\int_\Omega |\nabla u|^2 dx - \int_\Omega f u dx$$

- **Deep Ritz Method**:
  - Approximate u with a neural network u_θ.
  - Minimize $E(u_\theta)$ with respect to θ.

Approximation of the Integral

- **Challenge**:
  - Computing integrals in high dimensions
- **Solution**:
  - Use Monte Carlo integration:
    - Sample points $\{x_i\}$ uniformly in Ω.
    - Approximate integrals as averages over these samples.

Theorems

CONTRIBUTIONS AND MAIN RESULT

We show that neural networks of growing architecture that are trained with respect to suitably penalised Dirichlet energies converge to the solution of the Dirichlet problem (1). This is our main contribution and proves the consistency of the deep Ritz method proposed by E and Yu (2018).

More precisely, let $(\Theta_n)_{n\in\mathbb{N}}$ be sets of parameters of neural networks that we assume to be universal approximators in $H_0^1(\Omega)$ for $n\to\infty$. Now we introduce the objective functions

$$L_n\colon \Theta_n \to \mathbb{R}, \quad \theta \mapsto \int_\Omega \left(\frac{1}{2}\left|\nabla u_\theta\right|^2 - f u_\theta\right) \mathrm{d}x + n \int_{\partial\Omega} u_\theta^2 \mathrm{d}s, \tag{2}$$

where u_θ is the network arising from the parameters θ and $f \in L^2(\Omega)$ is some right hand side.

Theorem 1. *Let $(\theta_n)_{n\in\mathbb{N}}$ be a sequence of quasi-minimisers of the objective functions, meaning*

$$L_n(\theta_n) \le \inf_{\theta\in\Theta_n} L_n(\theta) + \delta_n,$$

where $\delta_n \to 0$. Then $(u_{\theta_n})_{n\in\mathbb{N}}$ converges to the solution u of the Dirichlet problem (1), both weakly in $H^1(\Omega)$ and strongly in $L^2(\Omega)$.

Figure: DEEP RITZ REVISITED: Johannes Muller, Marius Zeinhofer

Theorems

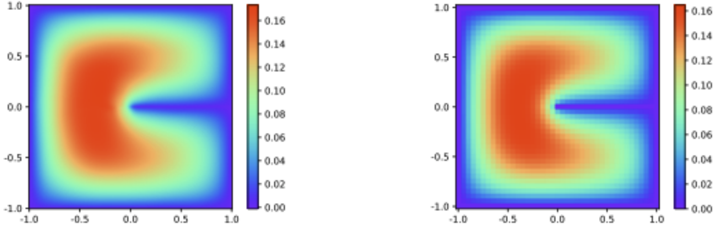
Theorem 3 *(i) Let u_P^* solve the Poisson equation (1) with $\|u_P^*\|_{\mathcal{B}(\Omega)} < \infty$. Let $u_{n,S}^m$ be the minimizer of the empirical loss $\mathcal{E}_{n,P}$ in the set $\mathcal{F} = \mathcal{F}_{\mathrm{SP}_\tau,m}(\|u_P^*\|_{\mathcal{B}(\Omega)})$ with $\tau = \sqrt{m}$. Then it holds that*

$$\mathbf{E}\big[\mathcal{E}_P(u_{n,P}^m) - \mathcal{E}_P(u_P^*)\big] \leq \frac{C_1\sqrt{m}(\sqrt{\log m}+1)}{\sqrt{n}} + \frac{C_2(\log m+1)^2}{m}. \tag{12}$$

Here $C_1 > 0$ depends polynomially on $\|u_P^*\|_{\mathcal{B}(\Omega)}, d, \|f\|_{L^\infty(\Omega)}$, and $C_2 > 0$ depends quadratically on $\|u_P^*\|_{\mathcal{B}(\Omega)}$.

Figure: A Priori Generalization Analysis of the Deep Ritz Method: Jianfeng, Yulong Lu, Min Wang

Experiments



Solution of Deep Ritz method, 811 parameters (b) Solution of finite difference method, 1,681 parameters

Figure: The Deep Ritz method: Weinan E, Bing Yu